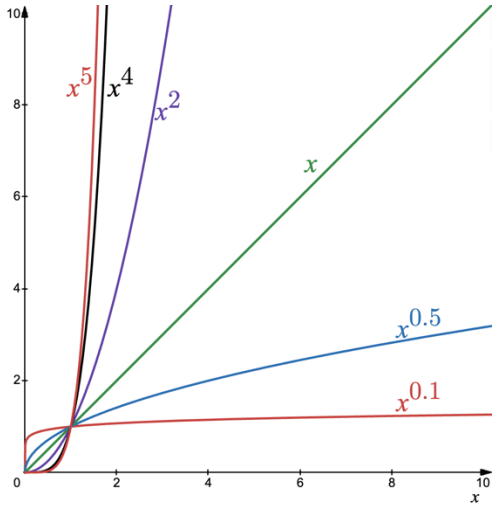
MAT1375, Classwork11, Fall2026

Ch10. Rational Functions II



6. The order of growth for functions: **Power functions**.

$$x^p, p > 0$$

For each $x > 1$, we have

$$\cdots < x^{0.1} < x^{0.5} < x < x^2 < x^4 < x^5 < \cdots$$

7. The definition of a **Horizontal Asymptote**:

The line $\boldsymbol{y = b}$ is a horizontal asymptote of the graph of a function f if $f(x)$ approaches $\boldsymbol{b}$ as $\boldsymbol{x}$ **increases** or **decreases without bound**.

8. What is the difference of Vertical Asymptote and Horizontal Asymptote?

9. How to locate Horizontal Asymptotes: Let $f(x) = \frac{p(x)}{q(x)}$ be a rational function given by

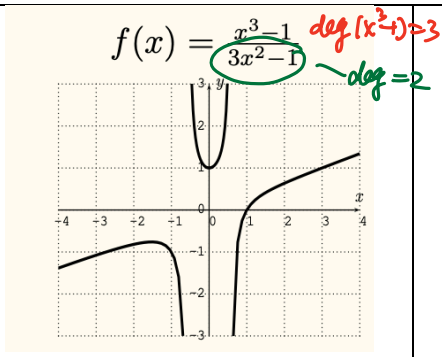
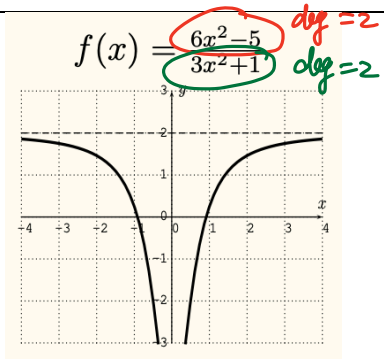
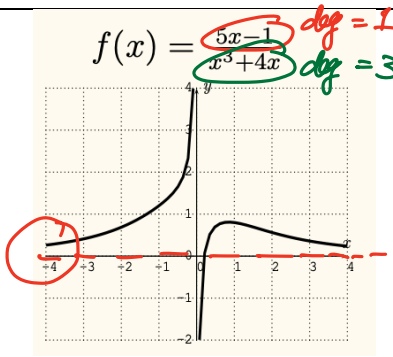
$$f(x) = \frac{p_n x^n + p_{n-1}x^{n-1} + \cdots + p_1 x + p_0}{q_m x^m + q_{m-1}x^{m-1} + \cdots + q_1 x + q_0}, p_n \neq 0, q_m \neq 0.$$

where the degree of the numerator is _____ and the degree of the denominator is _____.

(1) If $n > m$, the graph of f has _____ horizontal asymptote.

(2) If $n = m$, the line _____ (which is the ration of two _____) is the horizontal asymptote of the graph of f.

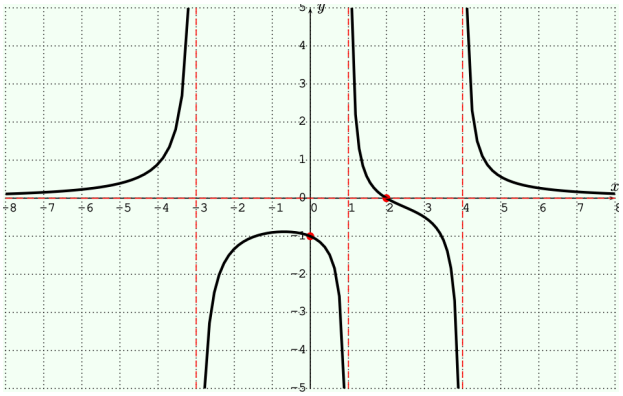
(3) If $n < m$, the _____ (which is_____) is the horizontal asymptote of the graph of f.

$f(x) = \frac{x^3-1}{3x^2-1}$ 	$f(x) = \frac{6x^2-5}{3x^2+1}$ 	$f(x) = \frac{5x-1}{x^3+4x}$ 
$\deg(p(x)) > \deg(q(x))$	$\deg(p(x)) = \deg(q(x))$	$\deg(p(x)) < \deg(q(x))$
_____	_____	_____

Ch11. More on Rational Functions

1. The graph of $f(x) = \frac{p(x)}{g(x)}$ is displayed below, where deg($p(x)$) =1 and deg($g(x)$)=3.

Find the intercepts, asymptotes, and a formula for $f(x)$.



For 2. And 3., let $f(x) = \frac{p(x)}{g(x)}$ be a rational function and **deg($p(x)$) > deg($g(x)$)**.

2. Rational Function and Long Division:

   If $p(x)$ divided by $g(x)$ can be represented with a quotient $q(x)$ and a remainder $r(x)$ where deg($r(x)$) ___ deg($g(x)$)**,** one can rewrite $f(x)$ as

$$f(x) = \frac{p(x)}{g(x)} = _____ + \frac{r(x)}{g(x)}.$$

3. Asymptotic Behavior with Slant Asymptote:

   Since deg($r(x)$) < deg($g(x)$), for large $|x|$ (which is $x \to \pm$ _____), we have

$$\frac{r(x)}{g(x)} \text{ approaches } _____ \text{ so that } f(x) _____ q(x).$$

   If $q(x)$ is a <u>linear</u> function (which is a polynomial of degree _____), then q is called the _____ asymptote of f.

4. Find the **slant asymptote** of the rational function $f(x) = \frac{2x^3-13x^2+35x-26}{x^2-4x+6}$.